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SECOND SEMESTER M.TECH EXAMINATION, JUNE–2019

GRAIN PROCESSING AND BAKING TECHNOLOGY

Time: 3 Hours **Maximum Marks:100**

Instructions: **i. Answer FIVE full questions. Question No. 1 and 2 are compulsory.**
ii. Draw neat sketches wherever necessary.
iii. Make suitable assumptions wherever necessary.

1.	**a.**	Recall the classification of cereal grains in India and the importance of cereals in human diet.	**10**
	b.	Illustrate the structure of maize grain in detail.	**10**
2.	**a.**	Summarize the milling process of wheat with neat flow diagram and the equipment involved.	**10**
	b.	List out the methods of parboiling and explain in detail about CFTRI and Jadavpur University methods.	**10**
3.		What do you mean by "Novel high fibre and whole grain breads"? Infer how that would effect the formulation and processing of cereal grains.	**20**
		OR	
4.		Elaborate on "Isolating and concentrating oat β-glucans using different methods".	**20**
5.	**a.**	Illustrate the layout of a bakery unit with prospects and problems, extend the process flow of biscuit manufacturing.	**10**
	b.	Inspect the process of dough rheology and explain the working principle behind the instruments used.	**10**
		OR	
6.	**a.**	List out the ingredients used for the production of cakes and analyze the methods involved for cake manufacturing.	**10**
	b.	Enlist the role of various physiochemical properties on bakery products, with example.	**10**
7.		Critically discuss the fortification and types of fortifications that are done in bakery industry.	**20**
		OR	
8.	**a.**	Investigate the significance of enriching soy and omega-3 of bread in current food habits of youth.	**10**
	b.	Examine the application of glycemic index and glycemic load on booming diabetic scenario.	**10**

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